

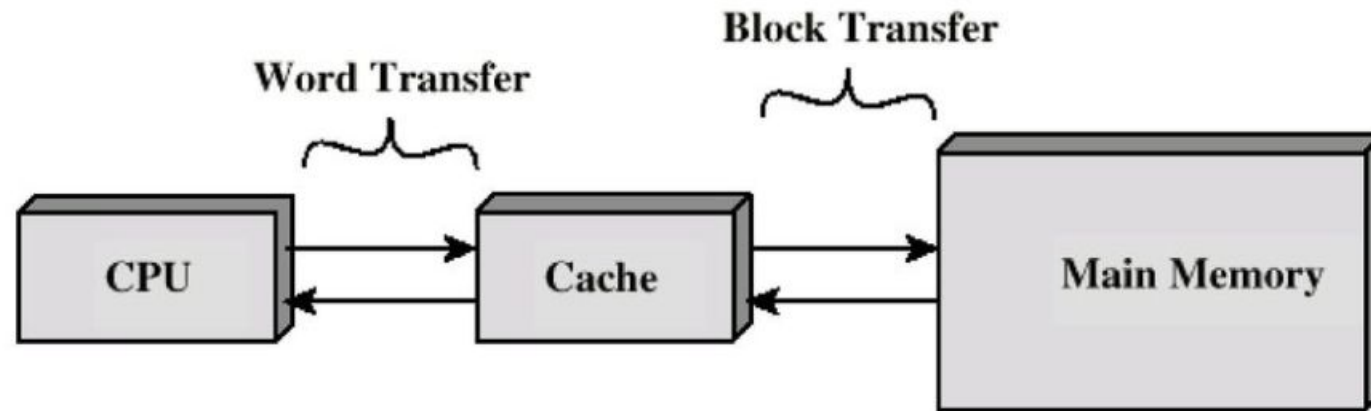
Cache Memory Mapping

Hierarchy List

- Registers
- L1 Cache
- L2 Cache
- Main memory
- Disk cache
- Disk
- Optical
- Tape

Cache

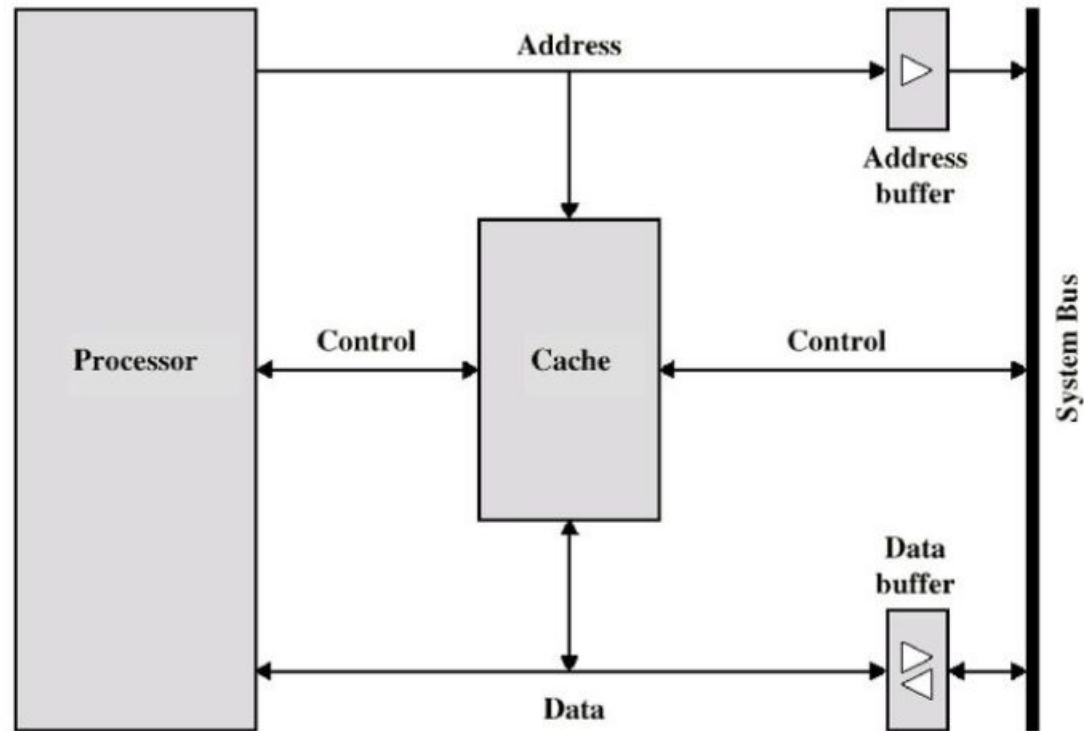
- Small amount of fast memory
- Sits between normal main memory and CPU
- May be located on CPU chip or module

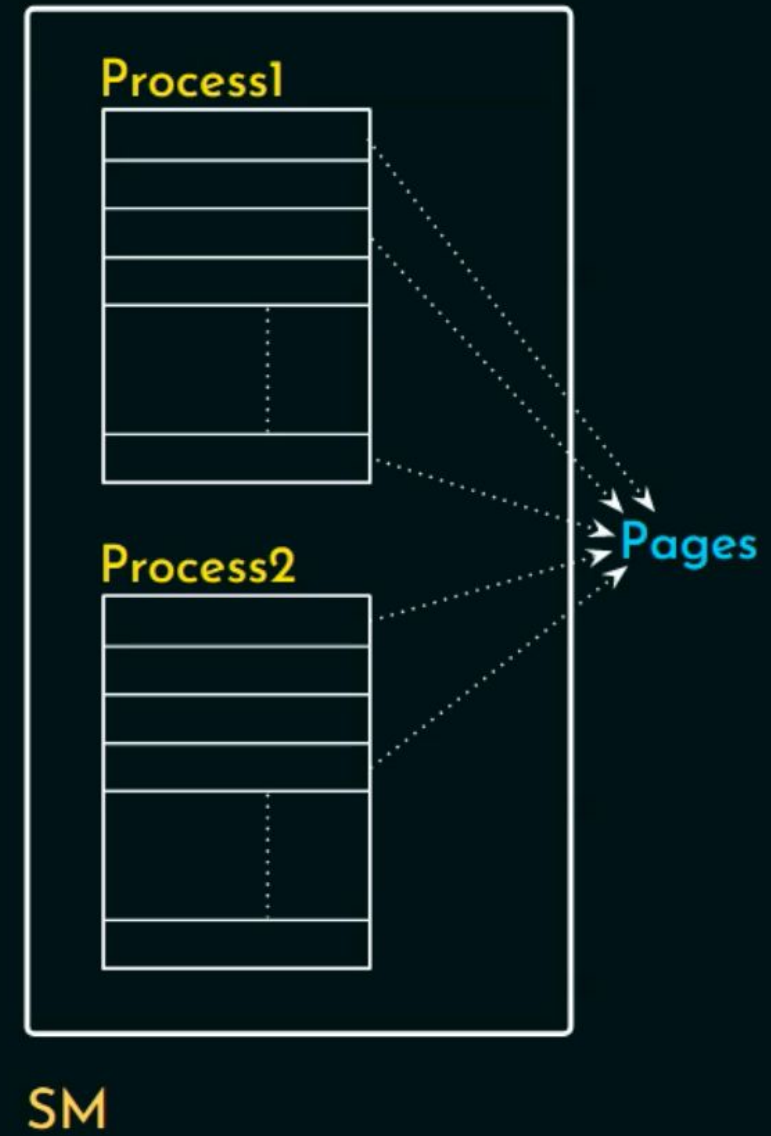
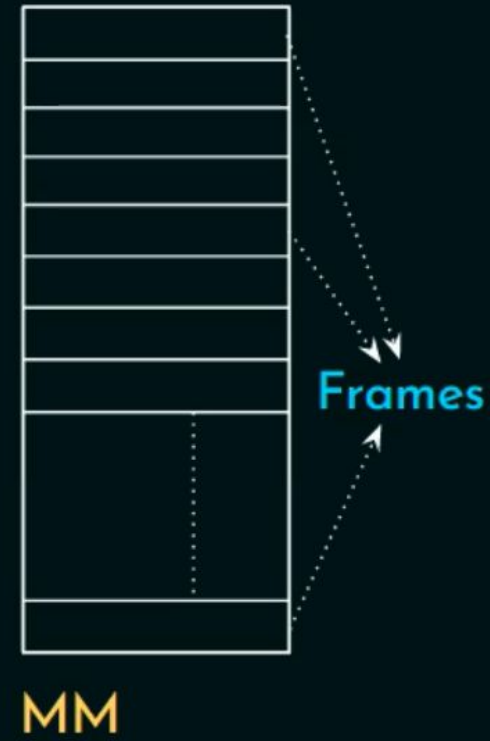
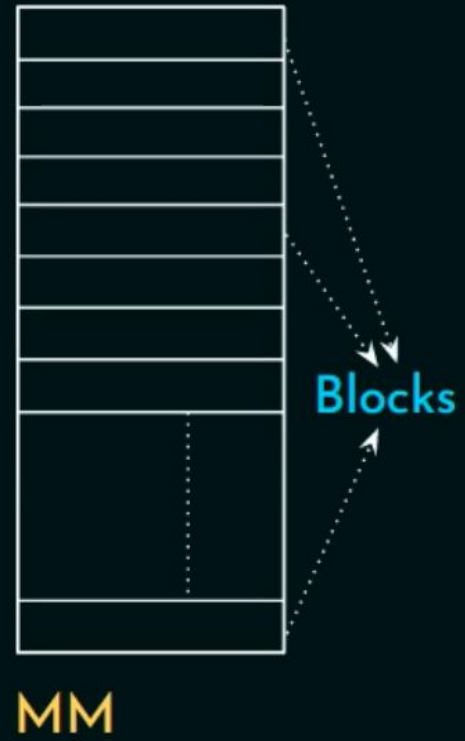
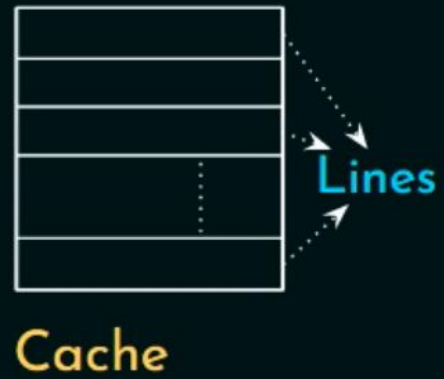


Cache Design

- Size
- Mapping Function
- Replacement Algorithm
- Write Policy
- Block Size
- Number of Caches

Typical Cache Organization





Line Size = Block Size

Frame Size = Page Size

Let's see an example

Main Memory Size : 64 words

Block Size : 4 words

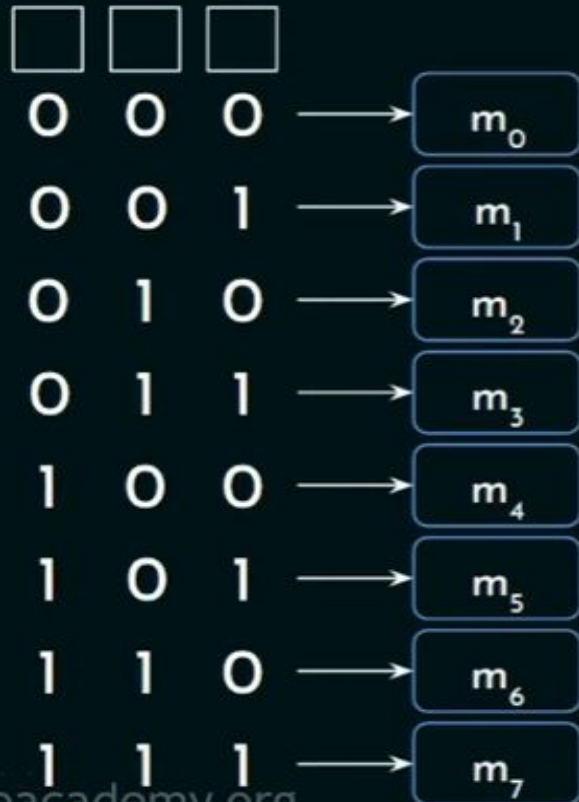
No. of Blocks in MM : $64 / 4 = 16$ i.e. 0, 1, 2, 3, , 15

0				
1				
2				
3				
4				
15				

Main Memory Size : 64 words i.e. (0, 1, ..., 63)

Block Size : 4 words

No. of Blocks in MM : $64 / 4 = 16$



64 words

$$\log_2 64 = \log_2 2^6 \\ = 6 \text{ bits}$$

0	0	1	2	3
1	4	5	6	7
2	8	9	10	11
3	12	13	14	15
4	16	17	18	19
5	20	21	22	23
6	24	25	26	27
7	28	29	30	31
8	32	33	34	35
9	36	37	38	39
10	40	41	42	43
11	44	45	46	47
12	48	49	50	51
13	52	53	54	55
14	56	57	58	59
15	60	61	62	63

Calculation: Physical Address Bits

0	1	2	3	0
4	5	6	7	1
8	9	10	11	2
12	13	14	15	3
16	17	18	19	4
60	61	62	63	15

Physical Address Space

16 Blocks

$$\log_2 16 = \log_2 2^4 \\ = 4 \text{ bits}$$

$$\log_2 64 = \log_2 2^6 = 6 \text{ bits}$$

P. A. bits :
(Physical Address bits)



0	0	→	0 th
0	1	→	1 st
1	0	→	2 nd
1	1	→	3 rd

Calculation: Block/Line size of Cache

Cache Size : 16 words

Line Size : 4 words

Block Size : 4 words

Block Size = Line Size

No. of Lines in Cache : $16 / 4 = 4$ i.e. 0, 1, 2, 3

4 Lines

$$\log_2 4 = \log_2 2^2$$

$$= 2 \text{ bits}$$



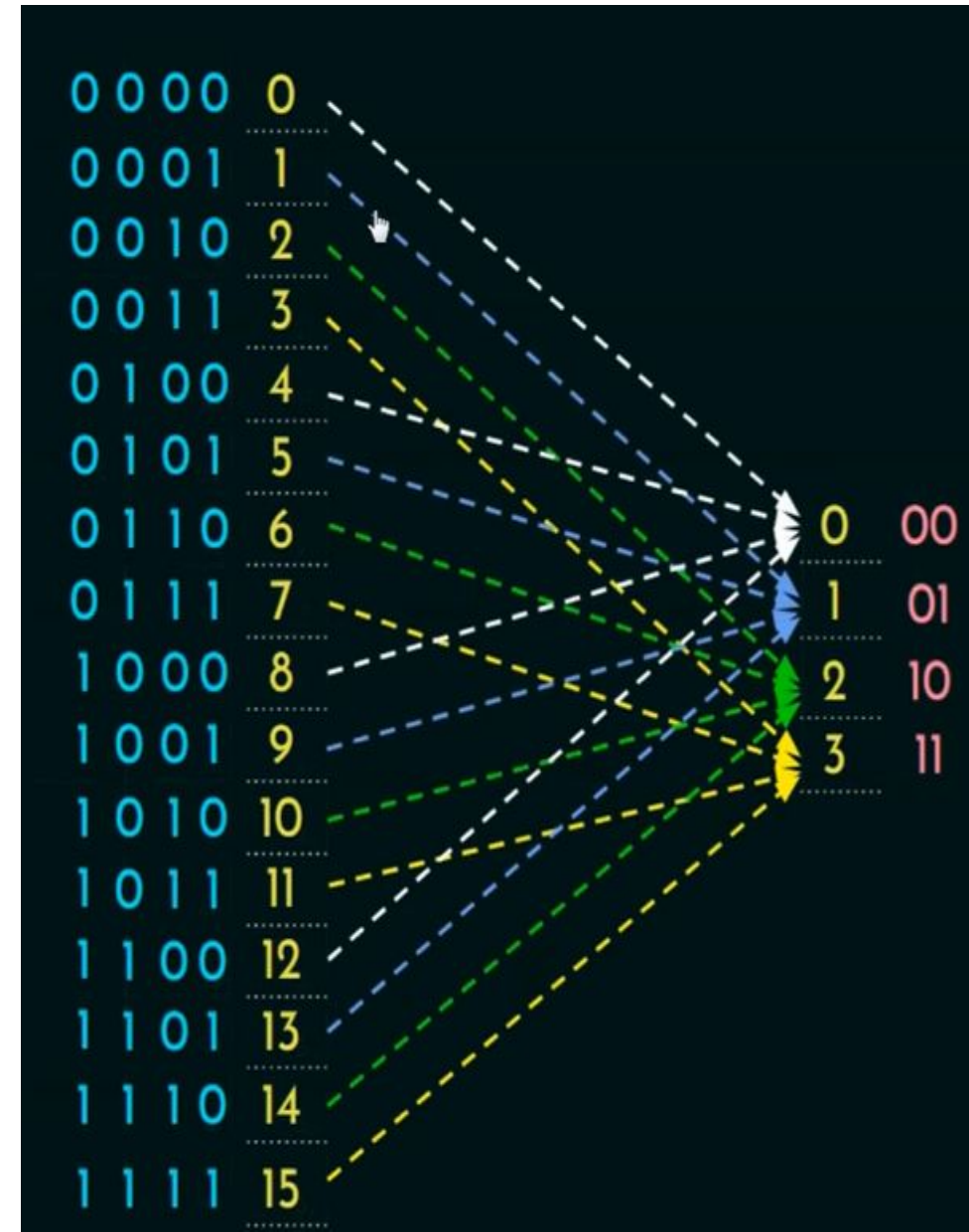
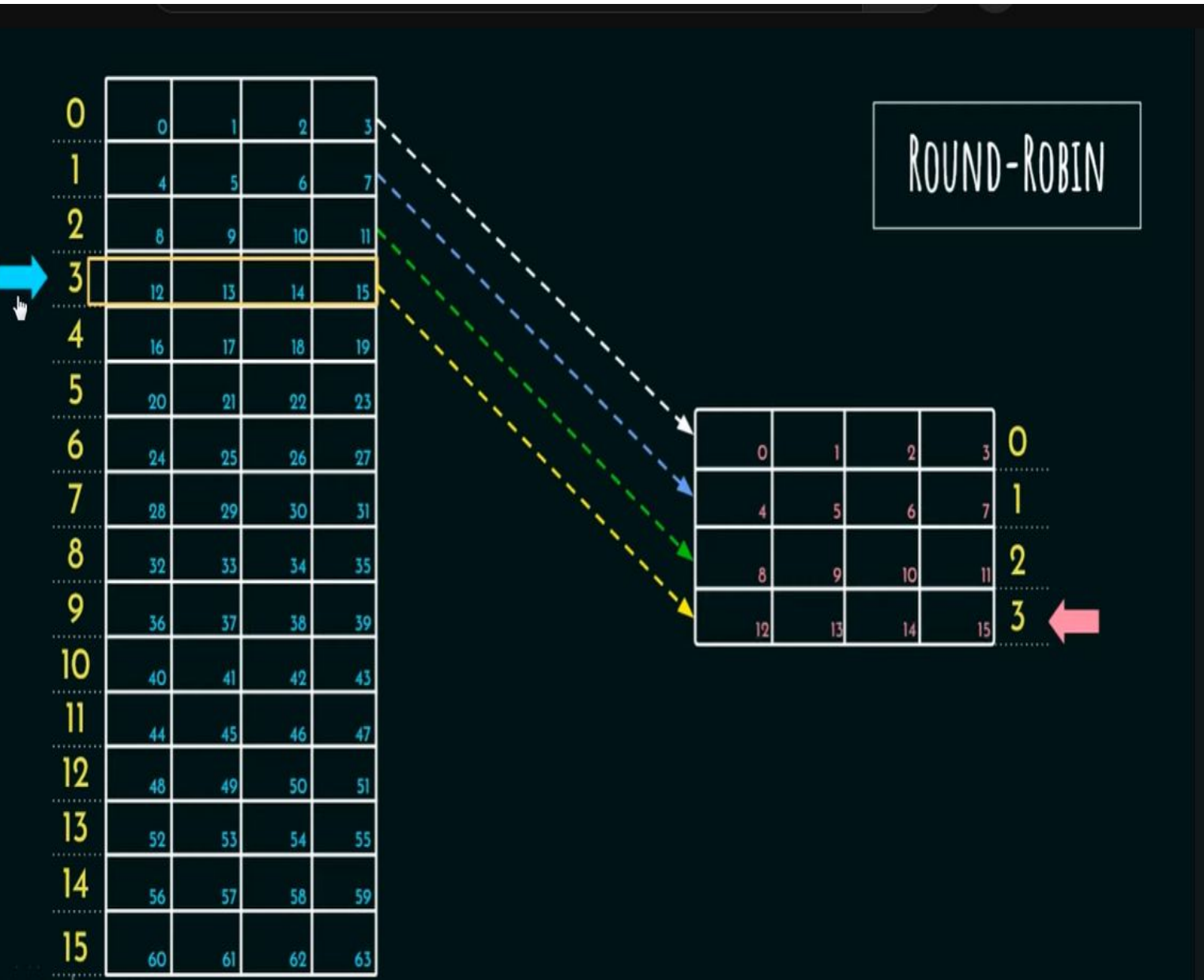
0 0 → 0

0 1 → 1

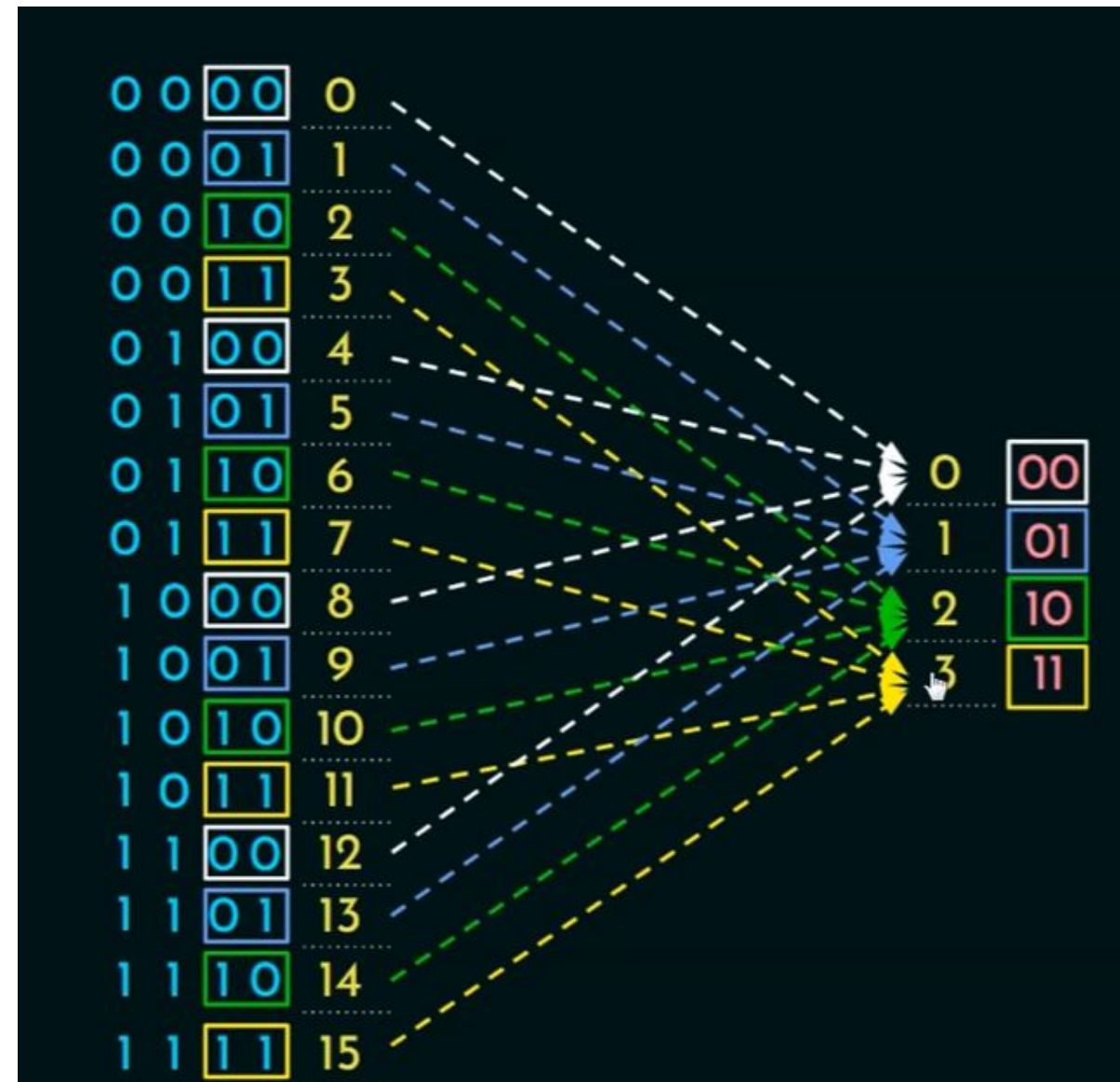
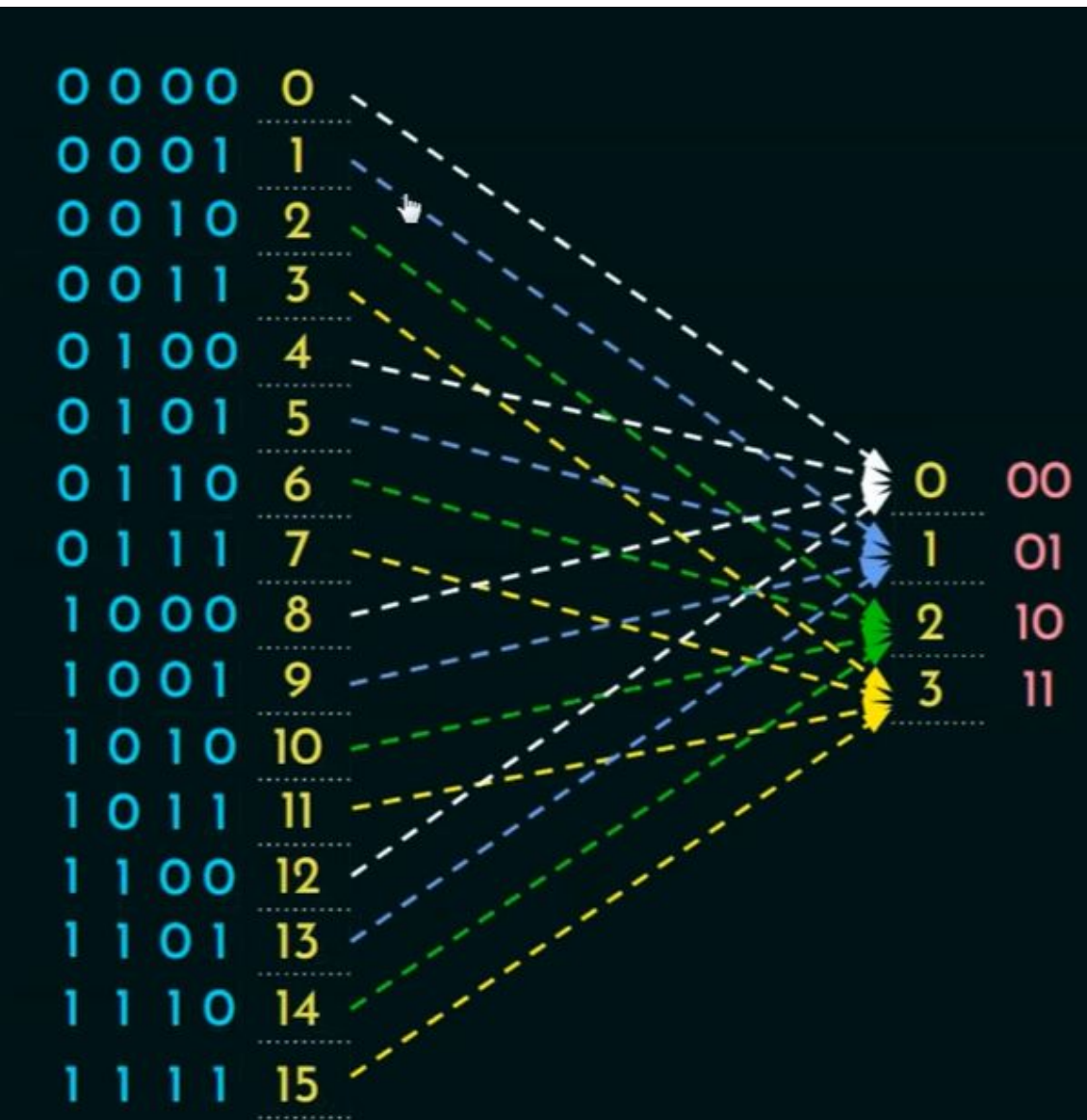
1 0 → 2

1 1 → 3

Main Memory to Cache Mapping



Complete Direct Mapping: MM to Cache



Address Structure: Direct Mapping

P. A. bits' split :

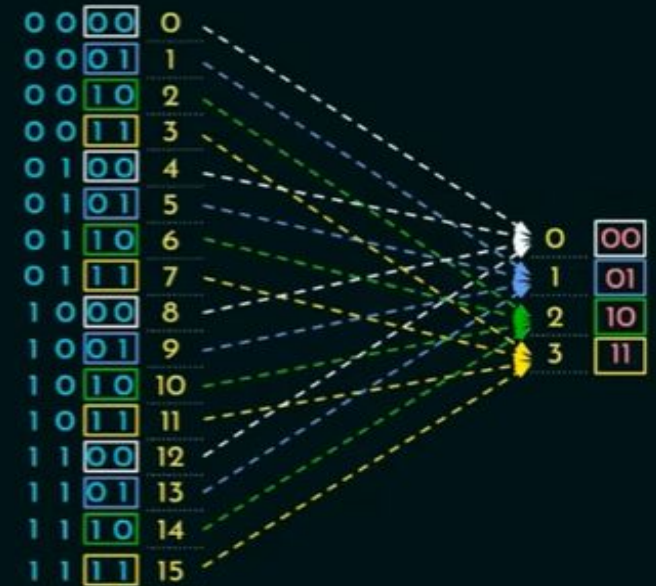
Block Number

Block / Line offset

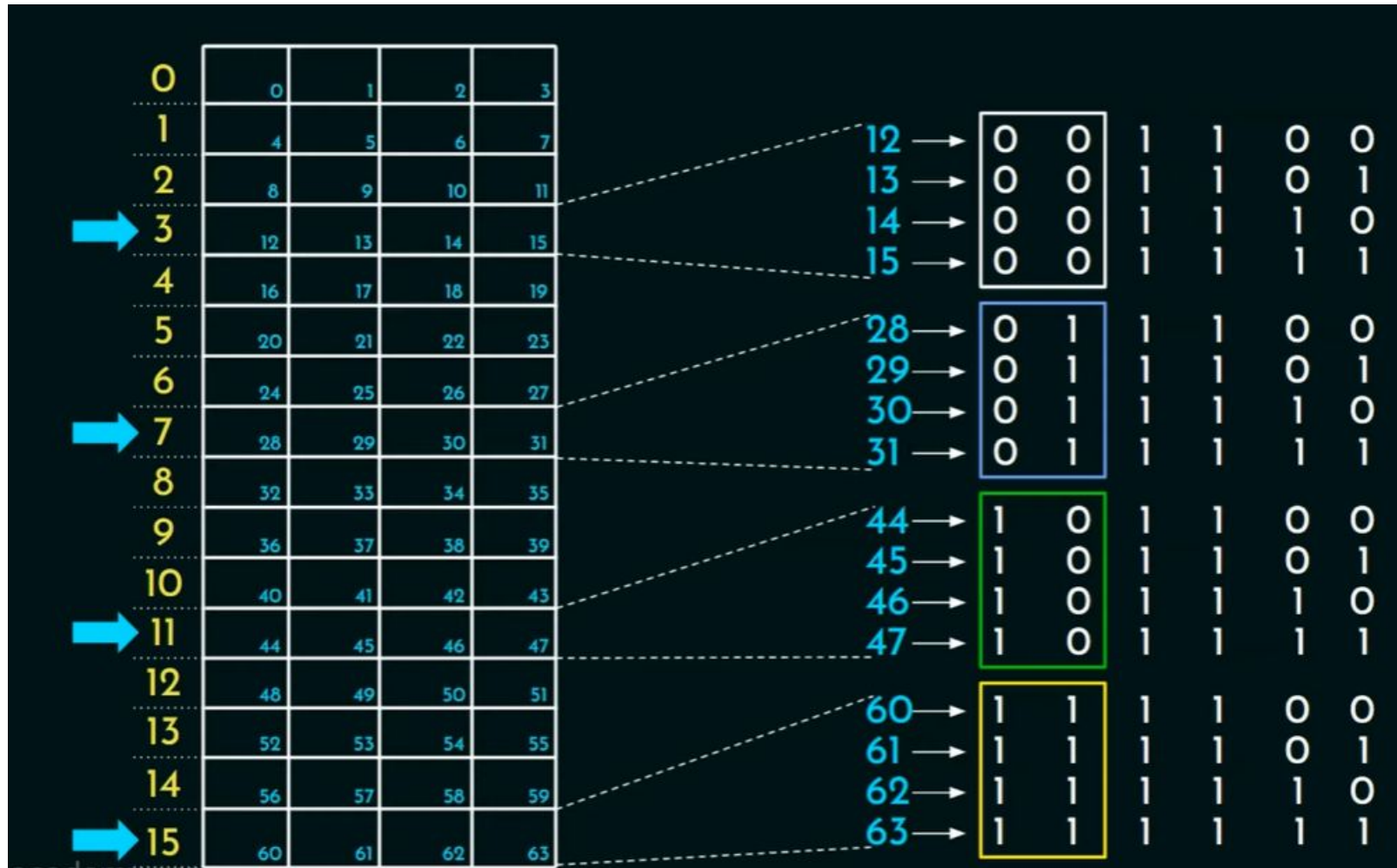
Tag bits

Line Number

Tag bits!?



How does the mapping work?



Let's try one example

Ex 1: MM Size: 4 GB
Cache Size: 1 MB
Block Size: 4 KB
Word Size: 1 Byte

Byte addressable memory

1. P.A. bits' split?
2. Tag directory size?

Sol. MM Size = 4 GB

1 Byte	=	8 bits
1 KB	=	1024 B(ytes) = 2^{10} B
1 MB	=	1024 KB = 2^{20} B
1 GB	=	1024 MB = 2^{30} B

Solution : Physical Address Bits' Split

Sol. MM Size = 4 GB = $2^2 \times 2^{30}$ B = $2^{(2+30)}$ B = 2^{32} B

$\therefore$ No. of P.A. bits = $\log_2 2^{32} = 32$

Block Size = 4 KB = $2^2 \times 2^{10}$ B = 2^{12} B

No. of **Blocks** in MM = $2^{32} / 2^{12} = 2^{20}$

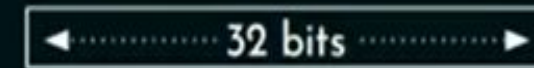
$\therefore$ **Block number** bits = $\log_2 2^{20} = 20$

Cache Size = 1 MB = 1×2^{20} B = 2^{20} B

No. of **Lines** in Cache = $2^{20} / 2^{12} = 2^8$

$\therefore$ **Line number** bits = $\log_2 2^8 = 8$

No. of **Tag** bits : P.A. bits - (Line no. bits + offset) = $32 - (8 + 12) = 12$



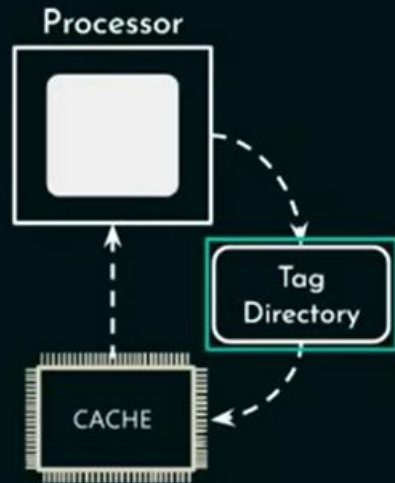
$\therefore$ **Block offset** = $\log_2 2^{12} = 12$



Tag Directory Size

Ex 1: MM Size: 4 GB
Cache Size: 1 MB
Block Size: 4 KB

Sol.



1. P.A. bits' split?

2. Tag directory size?

Tag directory:

- Keeps primarily the record of **Tag bits**, cache line-wise.
- No. of entries = No. of **Cache lines**.

32 bits		
20 bits		
Tag	Line	B/L offset
12 bits	8 bits	12 bits

No. of **Lines** in Cache = 2^8

No. of **Tag** bits = 12

Tag directory size = $2^8 \times 12$ bits = 3072 bits

Solve it!

Mapping Function

- Cache of 64kByte
- Cache block of 4 bytes
 - i.e. cache is 16k (2^{14}) lines of 4 bytes
- 16MBytes main memory
- 24 bit address
 - ($2^{24}=16\text{M}$)

Direct Mapping

- Each block of main memory maps to only one cache line
 - i.e. if a block is in cache, it must be in one specific place
- Address is in two parts
- Least Significant w bits identify unique word
- Most Significant s bits specify one memory block
- The MSBs are split into a cache line field r and a tag of $s-r$ (most significant)

Sample Solution

Direct Mapping Address Structure

Tag s-r	Line or Slot r	Word w
8	14	2

- 24 bit address
- 2 bit word identifier (4 byte block)
- 22 bit block identifier
 - 8 bit tag (=22-14)
 - 14 bit slot or line
- No two blocks in the same line have the same Tag field
- Check contents of cache by finding line and checking Tag

Direct Mapping Summary

- Address length = $(s + w)$ bits
- Number of addressable units = 2^{s+w} words or bytes
- Block size = line size = 2^w words or bytes
- Number of blocks in main memory = $2^{s+w}/2^w = 2^s$
- Number of lines in cache = $m = 2^r$
- Size of tag = $(s - r)$ bits

Cache Hit and Cache Miss

- If the processor finds that the memory location is in the cache, a [Cache Hit](#) has occurred and data is read from the cache.
- If the processor does not find the memory location in the cache, a **cache miss** has occurred. For a cache miss, the cache allocates a new entry and copies in data from the main memory, then the request is fulfilled from the contents of the cache.
- The formula for calculating a cache hit ratio, miss ratio is as follows:

`Hit Ratio(H) = hit / (hit + miss) = no. of hits/total accesses`

`Miss Ratio = miss / (hit + miss) = no. of miss/total accesses = 1 - hit ratio(H)`

Questions?

